Resolution Geometry: Source-Side Automation

Joint-Law Descent, Certified Correspondences, and a Fail-Closed H1–L1 HDF5 Audit

Matěj Rada
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Abstract

We finalize a conditional automation bridge, not an unconditional physical source reconstruction. The source-side object is the kernel groupoid of a declared observation law; an automated task is single-valued exactly when it is constant on its fibres. Otherwise the bridge returns a correspondence and does not choose a source representative. We distinguish a tuple of marginal laws from a genuinely coupled detector law, prove a confidence-set version of the bridge, and separate observable descent from covariance-dependent latent lifting. The supplied H1 and L1 binaries have the same GPS start, 1368993792, duration 4096 s, and sampling rate 16384 Hz. Their apparent text suffixes do not change their HDF5 signatures. Crucially, both injection masks equal 23 throughout: the supplied `NO_CW_HW_INJ` bit is unset. Under the explicitly declared all-five-injection-veto policy, zero windows are accepted. Descriptive computations on all 1024 finite four-second windows give relative covariance ranks 8, 7, and 15 at 10^{-3} for H1, L1, and the paired 18-feature map. These are not source Fisher ranks. Fourteen regression and counterexample modules pass their stated checks, including exact finite-law descent, dispatcher enumeration, and real-feature mutations. The result closes the automation contract while retaining the physical source law, response-aware transport, and statistical certificates as unsupplied inputs.

1 Scope and the meaning of a lock

The mature framework separates top-down structure forced by an observation law from bottom-up declarations of symmetry, task, nuisance, anchors, and latent covariance [1, 2]. This separation is decisive for automation. A deterministic program can execute a declared estimator without establishing that the estimator identifies a physical source. Likewise, a passing feature-rank test does not prove a source-side kernel or completeness statement.

This paper uses three distinct levels of completion:

1. **Record-side execution:** binary ingestion, provenance, metadata consistency, GPS pairing, quality masks, a fully specified feature recipe, and reproducible diagnostics.
2. **Conditional mathematical bridge:** exact task descent and set-valued inference for a supplied observation law and valid certificates.
3. **Physical source certification:** a calibrated source-response/noise model, a declared task, and accuracy and identifiability certificates. This level is not supplied by the attached strain records alone.

The first is executed here. The second is proved and has an exact finite-law reference adapter. The third is not certified. "Final" therefore means a closed contract with explicit abstention, not a claim that every requested source reconstruction succeeds.

The earlier placement paper [4] reported detector-local ranks nine and treated its records as unmatched. Those statements are historical reports about that computation, not transferable

certificates for the present binaries. We use the current files' metadata and a new, fully specified estimator. No claim of reproducing its undocumented estimator conventions is made.

2 The source-side object and the joint-law distinction

Definition 2.1 (Declared experiment)**.** *Let X be a declared latent domain, Y a record space, and $P : X \to \mathcal{P}(Y)$ an observation-law map. Deterministic protocols are included by taking $P_x = \delta_{F(x)}$. The kernel groupoid is*

$$\mathcal{K}_P = X \times_{\mathcal{P}(Y)} X, \qquad x \longrightarrow x' \quad \Longleftrightarrow \quad P_x = P_{x'}.$$

Composition is $(x, x') \circ (x', x'') = (x, x'')$. Its orbit space is $Q_P = X/\mathcal{K}_P \simeq \operatorname{im} P$ as a set.

The groupoid laws follow from equality. No metric, differentiable structure, physical unit, or preferred representative is implied. A declared groupoid A is sound when $A \subseteq \mathcal{K}_P$ and complete when its orbits exhaust the $\mathcal{K}_P$ fibres. Numerical rank is neither of these statements [1, 2].

Proposition 2.2 (Marginal tuple versus coupled law)**.** *For $M(x) = (P_x^H, P_x^L)$,*

$$\mathcal{K}_M = \mathcal{K}_{P^H} \cap \mathcal{K}_{P^L}.$$

For a coupled law P_x^{HL} with these marginals,

$$\mathcal{K}_{P^{HL}} \subseteq \mathcal{K}_M,$$

and inclusion can be strict. Equality holds if the coupling is a fixed function of the two marginals, in particular under declared conditional independence $P_x^{HL} = P_x^H \otimes P_x^L$.

Proof. Equality in a product is equality of each coordinate, proving the first statement. Marginalization preserves equality, proving inclusion. For strictness, the laws assigning mass $1/2$ to $(0,0), (1,1)$ and to $(0,1), (1,0)$ have the same Bernoulli marginals but different joint laws. If a fixed coupling rule reconstructs the joint law from the marginals, equality of marginals implies equality of joint laws. □

This corrects a potentially consequential ambiguity in the phrase "joint law." The product-valued law map used in the RG ladder is not automatically a dependence-aware probability law on simultaneous detector records. Coupling is an additional protocol when dependence is to carry source information.

For a regular conditionally independent likelihood, scores add and the cross-score expectation vanishes, so Fisher information adds. Without that factorization, neither Fisher additivity nor a product likelihood is licensed. GPS pairing supplies a time coordinate for coupled records; it does not establish conditional independence, common-source content, or the law P_x^{HL}.

3 The automation bridge

Theorem 3.1 (Exact task descent)**.** *For a task $T : X \to Z$, a unique map $\bar{T} : Q_P \to Z$ satisfying $T = \bar{T} \circ \pi$ exists if and only if*

$$P_x = P_{x'} \quad \Longrightarrow \quad T(x) = T(x'). \tag{1}$$

Equivalently, $\mathcal{K}_P \subseteq \mathcal{K}_T$. The assertion is set-theoretic; measurable or smooth descent requires the corresponding quotient hypotheses.

Proof. If $T = \bar{T} \circ \pi$, equal law fibres have equal quotient points and equal task values. Conversely define $\bar{T}([x]) = T(x)$. Condition (1) makes this well-defined; surjectivity of π makes it unique. □

This is the universal factorization theorem of the exact RG layer, applied to automation [1]. It is not a new physical identifiability theorem. It states exactly what a source-side program may compute without an extra choice.

Definition 3.2 (Maximal task correspondence)**.** *For a law $p \in \operatorname{im} P$, define*

$$\mathcal{T}(p) = \{T(x) : P_x = p\}.$$

If this set is not a singleton, the bridge returns the set rather than an arbitrary representative. A prior, regularizer, or covariance may select an estimate, but that estimate must retain its declaration and is not forced by p alone.

Theorem 3.3 (Confidence-set bridge)**.** *Suppose $C(y) \subseteq \mathcal{P}(Y)$ is a measurable confidence procedure satisfying*

$$\Pr_x\{P_x \in C(Y)\} \geq 1 - \alpha \qquad (x \in X).$$

Define $S(y) = \{x : P_x \in C(y)\}$ and $\mathcal{T}_C(y) = T(S(y))$. Then

$$\Pr_x\{T(x) \in \mathcal{T}_C(Y)\} \geq 1 - \alpha. \tag{2}$$

If an output $z(y)$ has a certified radius

$$\sup_{u \in S(y)} d_Z(z(y), T(u)) \leq \tau, \tag{3}$$

then $d_Z(z(Y), T(x)) \leq \tau$ with probability at least $1 - \alpha$.

Proof. On $P_x \in C(y)$, the true x belongs to $S(y)$ and hence its task belongs to the image. On the same event, (3) applies to $u = x$. □

A diameter bound $\operatorname{diam} T(S) > 2\tau$ rules out a radius-τ answer by the triangle inequality. The converse is not valid in arbitrary metric spaces: a centre and its radius still need certification. For scalar tasks on a bounded nonempty set, the midpoint of the infimum and supremum has radius half the diameter.

Equation (2) does not manufacture C. A finite trajectory does not certify an unrestricted law-space confidence ball of arbitrary strength. Dependence, regularity, model error, and sampling assumptions belong to the certificate. This is the same separation between samples and operator-level locks emphasized in the forgetting ladder [3].

Proposition 3.4 (Approximation and gate error)**.** *Suppose the implemented task differs uniformly from its exact target by at most η, and the implemented correspondence has certified radius τ_0. Then the exact task error is at most $\tau_0 + \eta$ on the coverage event. If additional certificate failures have probabilities at most α_g, the total failure probability is bounded by $\alpha + \sum_g \alpha_g$.*

Proof. Use the triangle inequality for the deterministic error and the union bound for the failure events. Independence is unnecessary. □

The HDF5 audit below does not provide such source-model approximation bounds. It therefore does not instantiate a physical version of this proposition.

4 What the groupoid does not automate

4.1 Approximate arrows do not form a kernel groupoid

The relation $d(P_x, P_{x'}) \leq \epsilon$ need not be transitive. On the real line, $0, 0.75, 1.5$ at $\epsilon = 1$ give two accepted adjacent pairs and a rejected endpoint pair. Taking a transitive closure can merge arbitrarily distant laws. The implementation must use exact law equality, a declared genuine partition, or confidence correspondences with explicit error accumulation. A thresholded covariance null space is not an exact groupoid fibre.

4.2 Observable transport is not a latent lift

Law-defined natural tensors descend on regular strata. A connection defined on the resulting observable geometry can be transported there under its own hypotheses. A source-side horizontal lift is additional structure. In a linear chart $J = [1\ 0]$, a covariance declaration $P > 0$ gives

$$B_P = PJ^\top (JPJ^\top)^{-1}, \qquad JB_P = 1.$$

For $P_1 = I$ and $P_2 = \left[\begin{smallmatrix} 1 & 1/2 \\ 1/2 & 1 \end{smallmatrix}\right]$, the lifts are $(1,0)^\top$ and $(1,1/2)^\top$. Both preserve the same observable increment and disagree latently. This is an explicit counterexample to deriving a unique latent lift from the observation law alone, consistent with the mature groupoid's separation of observable and declared latent transport [2, 1].

4.3 A conditional detector-response branch

A possible declared linear response model has the form

$$\widetilde{d}(f) = R(f;\lambda)\widetilde{s}(f) + \widetilde{n}(f),$$

where λ contains the declared response, timing, calibration, and nuisance choices. This equation is a proposed model interface, not a fitted result of this audit. If R is known, has full column rank, and its smallest singular value is bounded below by $\gamma > 0$, then $R^\dagger$ has norm at most $1/\gamma$. For a noiseless model $d = Rs$ with perturbation e,

$$\|R^\dagger(d+e) - s\| \le \|e\|/\gamma.$$

This elementary inequality is a conditional stability certificate. It does not identify an unknown R, remove response uncertainty, or reconstruct a source when the task varies along the joint-law fibre. With nuisance, the requested target and nuisance treatment must be specified before a Schur-complement Fisher metric can be used. The 18-feature covariance computed below is not R, DP, or a source Fisher matrix.

5 Executable contract and status discipline

The record runner and exact finite-law adapter have deliberately different scopes.

Record runner. It reads the binary container, hashes it, checks strain and mask clocks, records the mask names, applies the declared quality predicate, and computes feature diagnostics. Rejected windows are never silently relabeled as accepted. An empty accepted set produces no accepted-set covariance.

Exact finite-law adapter. The function `finite_source_bridge` accepts a finite table of exact rational probability rows and a target on its source indices. It validates probabilities, groups identical rows, and returns `LOCKED` only for a task constant on every fibre. Otherwise it returns `CORRESPONDENCE`. Its lock is explicitly scoped to exact finite-law task descent. It does not infer which law generated a single observed sample. This is a reference adapter, not a general continuous-law equality solver.

The dispatcher uses the mature status vocabulary [1]: `OUTSIDE_DOMAIN`, `INCONSISTENT`, `VACUOUS`, `ESCALATE`, `STRUCTURAL_WALL`, `CORRESPONDENCE`, and `LOCKED`. Domain and consistency precede task and vacuity; rank, candidate, and scale follow. Unknown gates cannot pass. The compact implementation enumerates 384 combinations of its declared flags and checks that the unique lock leaf is precisely their required conjunction. This is a control-flow regression, not a statistical calibration experiment.

A complete production source solver would additionally require machine-checkable source-law, accuracy, and confidence certificates. These are not implemented for the present physical records. The generic dispatcher is not wired to an invented source certifier. The audit returns `source_model=NOT_SUPPLIED` and `source_identification=OUTSIDE_DOMAIN`; separately, the strict injection gate is rejected. Multiple failure reasons are retained rather than hidden behind a favorable summary.

6 Actual HDF5 inputs and provenance

Both uploaded filenames end in `.hdf5.txt`, but their first eight bytes are the HDF5 signature `894844460d0a1a0a`; `h5py` opens both as binary HDF5. The input names are

```
H-H1_GWOSC_O4a_16KHZ_R1-1368993792-4096.hdf5.txt
L-L1_GWOSC_O4a_16KHZ_R1-1368993792-4096.hdf5.txt
```

The actual strain dataset is `strain/Strain`. The stored metadata and dataset attributes agree on the following values:

Quantity	H1	L1
GPS start	1368993792	1368993792
Duration (s)	4096	4096
Sampling rate (Hz)	16384	16384
Samples	67108864	67108864
Four-second windows	1024	1024
Finite primitive windows	1024	1024
Strictly accepted windows	0	0

The stored UTC start is `2023-05-24T20:02:54` in both. Equal GPS intervals justify contemporaneous record pairing. They do not assert a shared event or compensate propagation delay. The earlier unmatched-record distance correlation is not reused as a test on these paired files.

6.1 Quality and injection gate

For every second of a four-second window, the declared policy requires `DATA`, `CBC_CAT1`, and all five supplied no-hardware-injection bits. This conservative policy is a declaration for this audit, not a universal requirement for every scientific task.

H1 has DQ mask 511 for 3974 seconds and 415 for 122 seconds; L1 has 511 for 4092 seconds and 415 for four seconds. Both values retain the required `DATA` and `CBC_CAT1` bits. In contrast, each detector has injection mask 23 for all 4096 seconds. With the stored order

`NO_CBC_HW_INJ`, `NO_BURST_HW_INJ`, `NO_DETCHAR_HW_INJ`, `NO_CW_HW_INJ`, `NO_STOCH_HW_INJ`,

23 has bits 0,1,2,4 set and bit 3 unset. Thus the file does not assert the no-CW-hardware-injection condition. This is not an independent determination of an injection's waveform, strength, or astrophysical significance. It is sufficient to reject this declared all-five-bit gate. No accepted scientific sample remains under that policy.

The first development run exposed an empty-set covariance error because all windows were rejected. The runner was corrected to return no accepted-set spectrum and to label all-finite diagnostics separately. The policy was not weakened to make the rank computation pass. A mutation that drops the CW veto would accept mask 23 and is explicitly caught by the regression suite.

7 Fully specified descriptive feature experiment

For each raw four-second window h, let $z = h - \bar{h}$. The nine coordinates are natural logarithms of: population standard deviation $\sqrt{\langle z^2 \rangle}$; raw maximum absolute strain; Pearson kurtosis $\langle (z/\sigma)^4 \rangle$; and six band powers. The bands are

$$[20, 80), [80, 300), [300, 1000), [1000, 2048), [2048, 4096), [4096, 8192].$$

Band powers use the mean-subtracted full-window rectangular one-sided periodogram, with interior positive-frequency bins doubled and Nyquist not doubled. Summation multiplies by $\Delta f = f_s/N$. No taper, whitening, clipping, flooring, overlap, or resampling is used. A nonfinite or nonpositive primitive rejects that feature window. Kurtosis uses normalized residuals to avoid an unnecessary small-number fourth-power calculation.

The covariance is the sample covariance of centered log features with denominator $n - 1$, without marginal standardization. Relative rank counts eigenvalues strictly greater than $\epsilon\lambda_{\max}$. It is a descriptive scale convention, not a confidence-certified rank wall.

All-finite descriptive map	Dimension	Rank 10^{-3}	Rank 10^{-2}
H1	9	8	5
L1	9	7	3
GPS-paired concatenation	18	15	7

All rows use the 1024 finite windows, not a quality-accepted sample. The paired covariance condition number is approximately 9789.99. The relative eigenvalues are approximately

$$\begin{aligned}(&1, .645627, .058481, .024071, .019777, .015718,\\ &.013292, .009823, .007815, .004675, .004252, .003867,\\ &.003369, .001258, .001196, .000703, .000117, .000102).\end{aligned}$$

The closest eigenvalue to 10^{-3} has relative margin about 1.9611×10^{-4}. This is a numerical threshold margin, not a sampling confidence interval. Same-coordinate cross-detector feature correlations range from approximately -0.0666 to 0.0195. No iid significance test, event detection, or coherent waveform claim is attached to them.

The difference from the earlier rank-nine report is preserved rather than normalized away. Exact estimator and input provenance are required for a numerical comparison. A nonlinear derived coordinate can increase finite covariance rank without adding intrinsic source dimension; only affine duplication forces covariance singularity automatically. Likewise, full-span principal-angle agreement is not a nontrivial transport certificate [4].

8 Attacks, regressions, and their limits

The executable suite has fourteen modules. Each records its check and a named mutation or counterexample. Exact algebraic fixtures are not simulated detector data; they test mathematical and control-flow statements. Real-feature checks use the supplied strain records. No simulated strain or synthetic source population replaces them.

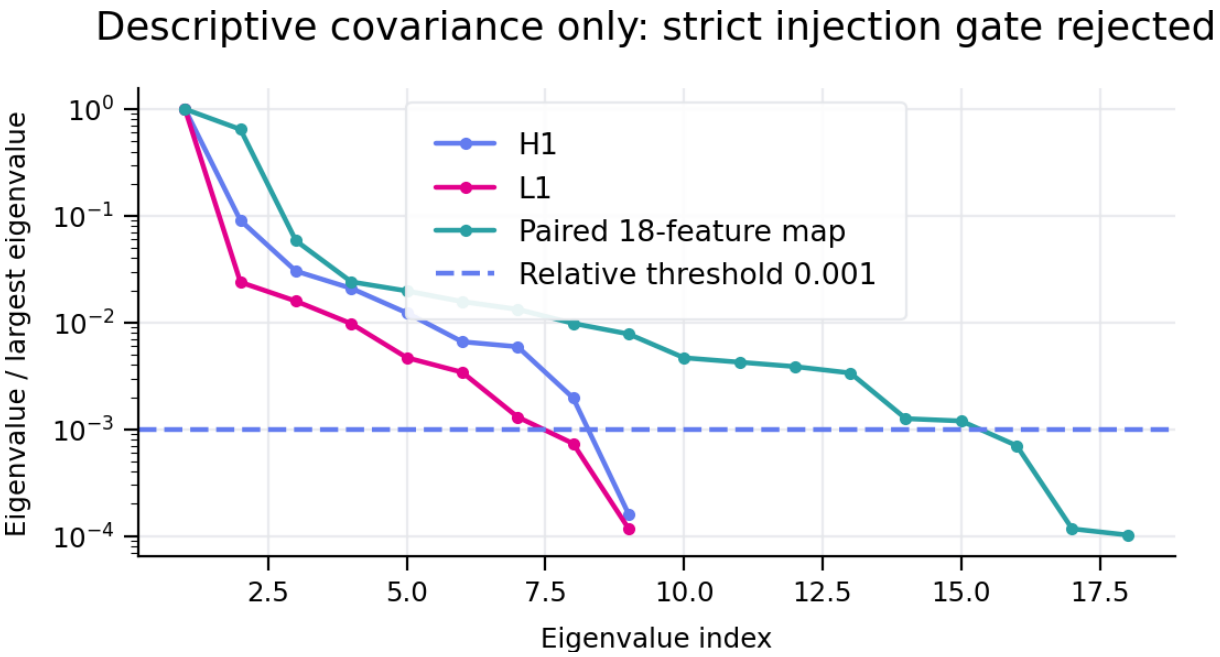


Figure 1: Actual all-finite feature covariance spectra. Every strict injection-quality window was rejected. The plot describes the records and cannot certify source identifiability or an exact null sector.

ID	Target	Mutation or counterexample
B01	Binary HDF5 provenance	Text bytes fail signature
B02	GPS interval pairing	One-second metadata shift
B03	Per-second DQ predicate	Clear DATA in one second
B04	Exact joint kernel intersection	Union replaces intersection
B05	Coupled law versus marginals	Equal marginals, opposite coupling
B06	Task descent on 256 finite cases	Nonconstant task on one fibre
B07	Tolerance is not transitive	Chain merges distant endpoints
B08	Real-feature chart permutation	Appended affine duplicate
B09	Deterministic post-map proxy	Perturb weights by one percent
B10	Dispatcher on 384 flag states	Force lock for multiple candidates
B11	Declared latent lift	Two covariances, same observation
B12	Failed source gate cannot lock	Substitute multiple-candidate lock
B13	Actual injection mask policy	Omit the CW no-injection bit
B14	Exact finite-law bridge adapter	Select first source representative

All fourteen checks and their recorded sensitivity conditions pass. Their evidential strengths differ. B04–B07 and B11 are exact counterexample or theorem-fixture tests. B08 is a real-feature numerical regression. B09 verifies an algebraic post-map identity and its sensitivity; it does not validate a physical energy interpretation. B10 and B12 are status-contract regressions, not attacks on an implemented physical inverse solver. B14 directly exercises the finite-law adapter, but does not certify a continuous source law.

In particular, the suite does not establish a false-lock frequency for physical sources, calibrate a confidence region, test a detector response model, or demonstrate out-of-sample source reconstruction. Its passing count must not be promoted into any of those claims. Proofs establish the conditional mathematics; the executable checks guard specific implementation and overclaim failures.

9 Final result and non-claims

The automation bridge closes in the following precise sense:

1. The source-side equivalence object is computed from a *declared full law*, not from a covariance rank or an inferred symmetry label.

2. A task is law-determined exactly when it descends through the kernel quotient. Otherwise the result is a correspondence.

3. Valid law confidence sets push through the bridge without losing coverage; point answers require an additional certified radius and numerical-accuracy budget.

4. Observable descent does not select a latent lift. A covariance, prior, response, or regularizer remains a declaration.

5. The supplied records are paired in GPS time, but fail the explicitly declared all-five-injection-veto gate. The program records this failure and still permits clearly labeled descriptive diagnostics.

There is no lock here for a unique astrophysical source, source waveform, polarization reconstruction, sky localization, physical stress-energy tensor, detector-independent geometry, exact source kernel, permanent declaration forgetting, or response-aware physical transport. The source model and the corresponding statistical certificates are absent, not implicitly supplied by the groupoid.

The strongest supported final statement is therefore: *the conditional source-automation contract is mathematically closed and its finite-law reference adapter is tested; the present H1–L1 execution correctly abstains from physical source certification.* A fail-closed result is an intended output of mature automation, not a reason to relax a gate after inspecting the data.

A Reproducibility and integrity

Run the supplied script with Python, NumPy, and h5py:

```
python rg_source_automation.py -input-dir .  -out-dir .
```

The input directory must contain exactly one matching H1 and L1 binary. The suffix may remain `.hdf5.txt`; the reader uses the binary content. The runner emits `rg_source_results.json` and the two feature CSVs, including GPS starts and strict acceptance flags. The JSON records estimator conventions, package versions, mask names and counts, spectra, test details, verdicts, and the script SHA-256. The CSVs retain all finite descriptive rows with acceptance flags zero.

Input SHA-256 digests are:

```
H1:  dc63726cee1066ecea9c1e1d6d7f7a3a
   728921a0b1047202250913e6c2c74d50

L1:  1e042a4ac352eb60ea12474f7155ed280
   25c9410efe679097d3510db247a6112
```

The digests are each the concatenation of their two displayed lines. No external web source was consulted. The mathematical context is the supplied working papers, and the empirical evidence is the attached binary records. External bibliographic or current scientific verification is outside this audit.

References

[1] M. Rada. *Resolution Geometry*. Supplied 24-page working paper, especially Sections 2–7, 10, and 14.

[2] M. Rada. *RG – The Groupoid of Resolution*. Supplied 8-page working paper, especially Levels 0–5 and 7.

[3] M. Rada. *RG – Uncertainty Geometry: The Forgetting Ladder*. Supplied October 2026 working paper, especially R30–R47 and the certificate/accuracy distinction.

[4] M. Rada. *Gauge-Free Observable-Sector Extraction, Detector-Local Admissibility, and the Limits of Cross-Detector Geometry Identity in LIGO O4a H1–L1 Strain Data*. Supplied July 30, 2026 working paper.